	School of Engineering & Technology	
	Department: SOET	**Session: 24-25 Even**
	Program: B. Tech CSE	**Semester: IV**
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	Course Name: Database Management Systems Lab	**Faculty: Ms. Mansi Kajal**

Lab Sheet: 1

Objective:

- To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

Instructions:

1. Open your preferred Database Management System (DBMS) tool.
2. Create a new database named DBMS_YourName.
3. Draw ER diagrams for the given scenarios.
4. Convert entities and relationships into relational schemas.
5. Create tables using the derived schemas.

Relational Schema

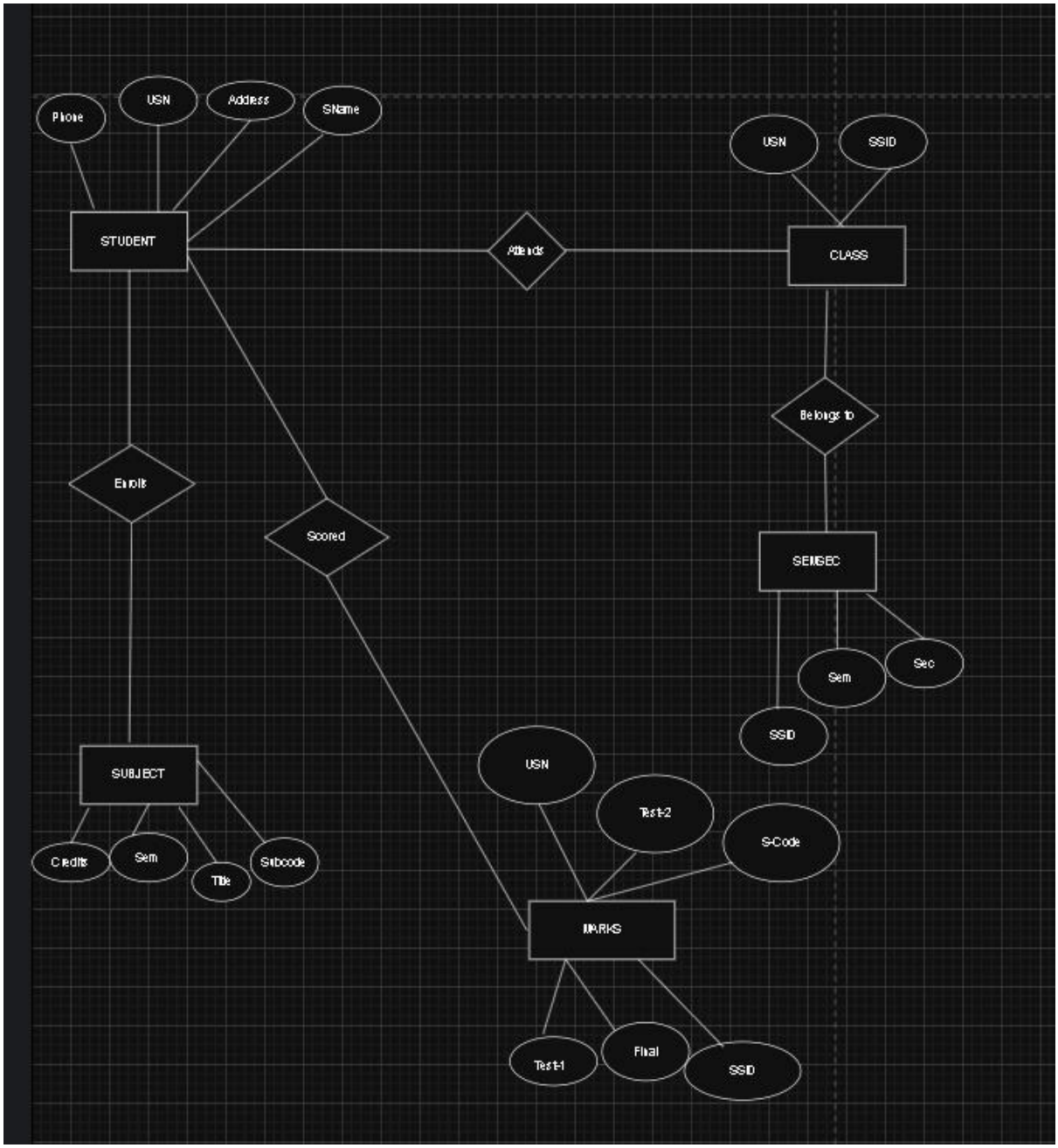
PART A – COLLEGE DATABASE

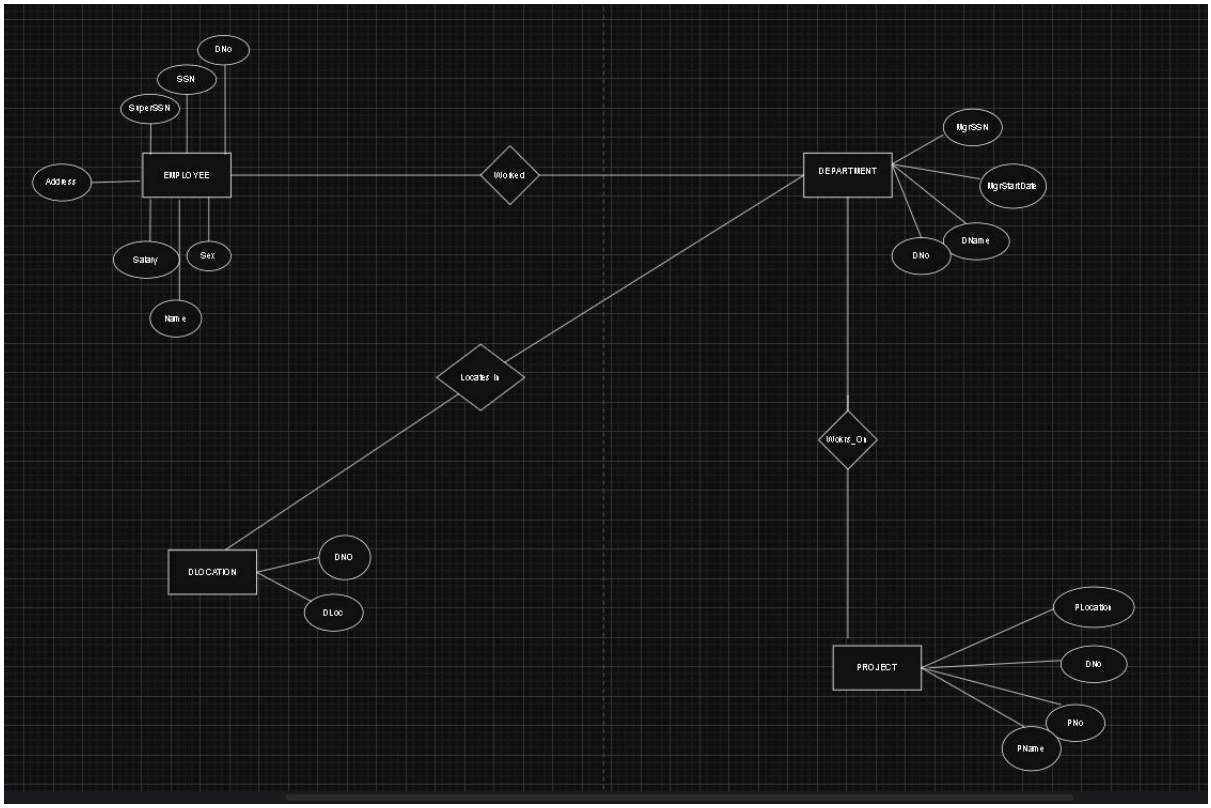
STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)
DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)
DLOCATION (Dno, Dloc)

	PROJECT (PNo, PName, PLocation, DNo) WORKS_ON (SSN, PNo, Hours) **PART C – CUSTOMER SALE SCHEMA** • Database Schema for a customer-sale scenario • Customer(Cust id : integer, cust_name: string) • Item(item_id: integer, item_name:string, price: integer) Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)


Phone
USN
Address
SName
STUDENT
Attends
USN
SSID
CLASS
Belongs to
SEMSEC
Sem
Sec
SSID
Enrolls
Scored
SUBJECT
Credits
Sem
Title
Subcode
USN
Test-2
S-Code
MARKS
Test-1
Final
SSID



DDL Questions (Data Definition Language):

1. Write SQL statements to create all the above relations with appropriate PRIMARY KEY and FOREIGN KEY constraints.
2. Add a new column Email to the STUDENT relation.
3. Apply NOT NULL constraints to Salary and Price attributes.
4. Rename attribute SName to StudentName in STUDENT relation.
5. Drop the DLOCATION relation.

```
Database changed
mysql> CREATE TABLE STUDENT (
    ->     USN VARCHAR(20) PRIMARY KEY,
    ->     SName VARCHAR(50),
    ->     Address VARCHAR(100),
    ->     Phone VARCHAR(15),
    ->     Gender VARCHAR(10)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE SEMSEC (
    ->     SSID VARCHAR(10) PRIMARY KEY,
    ->     Sem INT,
    ->     Sec VARCHAR(5)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE CLASS (
    ->     USN VARCHAR(20),
    ->     SSID VARCHAR(10),
    ->     PRIMARY KEY (USN, SSID),
    ->     FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    ->     FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
    -> );
ERROR 1050 (42S01): Table 'class' already exists
mysql>
mysql> CREATE TABLE SUBJECT (
    ->     Subcode VARCHAR(10) PRIMARY KEY,
    ->     Title VARCHAR(50),
    ->     Sem INT,
    ->     Credits INT
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> CREATE TABLE IAMARKS (
    ->     USN VARCHAR(20),
    ->     Subcode VARCHAR(10),
    ->     SSID VARCHAR(10),
    ->     Test1 INT,
    ->     Test2 INT,
    ->     Test3 INT,
    ->     FinalIA INT,
    ->     PRIMARY KEY (USN, Subcode, SSID),
    ->     FOREIGN KEY (USN) REFERENCES STUDENT(USN),
```

```
mysql> CREATE TABLE EMPLOYEE (
    ->     SSN VARCHAR(20) PRIMARY KEY,
    ->     Name VARCHAR(50),
    ->     Address VARCHAR(100),
    ->     Sex VARCHAR(10),
    ->     Salary DECIMAL(10,2),
    ->     Superssn VARCHAR(20),
    ->     Dno INT,
    ->     FOREIGN KEY (Superssn) REFERENCES EMPLOYEE(SSN)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE DEPARTMENT (
    ->     Dno INT PRIMARY KEY,
    ->     Dname VARCHAR(50),
    ->     Mgrssn VARCHAR(20),
    ->     Mgrstartdate DATE,
    ->     FOREIGN KEY (Mgrssn) REFERENCES EMPLOYEE(SSN)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE DLOCATION (
    ->     Dno INT,
    ->     Dloc VARCHAR(50),
    ->     PRIMARY KEY (Dno, Dloc),
    ->     FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE PROJECT (
    ->     PNo INT PRIMARY KEY,
    ->     PName VARCHAR(50),
    ->     PLocation VARCHAR(50),
    ->     DNo INT,
    ->     FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> CREATE TABLE WORKS_ON (
    ->     SSN VARCHAR(20),
    ->     PNo INT,
    ->     Hours DECIMAL(5,2),
    ->     PRIMARY KEY (SSN, PNo),
    ->     FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
```

```
mysql> CREATE TABLE CUSTOMER (
    ->     Cust_id INT PRIMARY KEY,
    ->     cust_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE ITEM (
    ->     item_id INT PRIMARY KEY,
    ->     item_name VARCHAR(50),
    ->     price INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE SALE (
    ->     bill_no INT,
    ->     bill_date DATE,
    ->     cust_id INT,
    ->     item_id INT,
    ->     qty_sold INT,
    ->     PRIMARY KEY (bill_no, item_id),
    ->     FOREIGN KEY (cust_id) REFERENCES CUSTOMER(Cust_id),
    ->     FOREIGN KEY (item_id) REFERENCES ITEM(item_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> ALTER TABLE STUDENT
    -> ADD Email VARCHAR(100);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE EMPLOYEE
    -> MODIFY Salary DECIMAL(10,2) NOT NULL;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> ALTER TABLE ITEM
    -> MODIFY price INT NOT NULL;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE STUDENT
    -> RENAME COLUMN SName TO StudentName;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

DML Questions (Data Manipulation Language):

1. Insert at least 10 tuples into CUSTOMER, ITEM, and SALE relations.
2. List all bills of the current date with customer names and item names.
3. Display total bill amount using quantity and price.
4. List customers who purchased products priced above 200.
5. Count the number of products purchased by each customer.
6. List products purchased by customer having Cust_id = 5.
7. Create a VIEW showing bill_no, bill_date, cust_id, item_id, price, qty,sold, amount.

Create a VIEW showing daily sales date-wise for the last one week.

```
mysql> INSERT INTO CUSTOMER VALUES
    -> (1,'Rahul'), (2,'Anita'), (3,'Vikram'),
    -> (4,'Sneha'), (5,'Arjun'), (6,'Meena'),
    -> (7,'Ravi'), (8,'Priya'), (9,'Kiran'), (10,'Asha');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO ITEM VALUES
    -> (1,'Pen',10), (2,'Notebook',50), (3,'Bag',800),
    -> (4,'Bottle',250), (5,'Shoes',1200),
    -> (6,'Shirt',700), (7,'Watch',1500),
    -> (8,'Mouse',400), (9,'Keyboard',900), (10,'Book',300);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO SALE VALUES
    -> (101,CURDATE(),1,3,1),
    -> (102,CURDATE(),2,5,2),
    -> (103,CURDATE(),5,4,1),
    -> (104,CURDATE(),5,7,1),
    -> (105,CURDATE(),3,2,5),
    -> (106,CURDATE(),4,8,2),
    -> (107,CURDATE(),6,9,1),
    -> (108,CURDATE(),7,6,1),
    -> (109,CURDATE(),8,10,3),
    -> (110,CURDATE(),9,1,10);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT S.bill_no, C.cust_name, I.item_name
    -> FROM SALE S
    -> JOIN CUSTOMER C ON S.cust_id = C.Cust_id
    -> JOIN ITEM I ON S.item_id = I.item_id
    -> WHERE S.bill_date = CURDATE();
+---------+-----------+-----------+
| bill_no | cust_name | item_name |
+---------+-----------+-----------+
|     101 | Rahul     | Bag       |
|     102 | Anita     | Shoes     |
|     103 | Arjun     | Bottle    |
|     104 | Arjun     | Watch     |
|     105 | Vikram    | Notebook  |
|     106 | Sneha     | Mouse     |
|     107 | Meena     | Keyboard  |
|     108 | Ravi      | Shirt     |
|     109 | Priya     | Book      |
|     110 | Kiran     | Pen       |
```

```
mysql> SELECT S.bill_no,
    ->        SUM(I.price * S.qty_sold) AS Total_Amount
    -> FROM SALE S
    -> JOIN ITEM I ON S.item_id = I.item_id
    -> GROUP BY S.bill_no;
+---------+--------------+
| bill_no | Total_Amount |
+---------+--------------+
|     101 |          800 |
|     102 |         2400 |
|     103 |          250 |
|     104 |         1500 |
|     105 |          250 |
|     106 |          800 |
|     107 |          900 |
|     108 |          700 |
|     109 |          900 |
|     110 |          100 |
+---------+--------------+
10 rows in set (0.01 sec)

mysql> SELECT DISTINCT C.cust_name
    -> FROM SALE S
    -> JOIN CUSTOMER C ON S.cust_id = C.Cust_id
    -> JOIN ITEM I ON S.item_id = I.item_id
    -> WHERE I.price > 200;
+-----------+
| cust_name |
+-----------+
| Rahul     |
| Arjun     |
| Anita     |
| Ravi      |
| Sneha     |
| Meena     |
| Priya     |
+-----------+
7 rows in set (0.01 sec)

mysql> SELECT C.cust_name, COUNT(S.item_id) AS Total_Products
    -> FROM SALE S
    -> JOIN CUSTOMER C ON S.cust_id = C.Cust_id
    -> GROUP BY C.cust_name;
+-----------+----------------+
| cust_name | Total_Products |
+-----------+----------------+
| Rahul     |              1 |
| Anita     |              1 |
| Vikram    |              1 |
| Sneha     |              1 |
```

```
mysql> SELECT I.item_name
    -> FROM SALE S
    -> JOIN ITEM I ON S.item_id = I.item_id
    -> WHERE S.cust_id = 5;
+-----------+
| item_name |
+-----------+
| Bottle    |
| Watch     |
+-----------+
2 rows in set (0.00 sec)

mysql> CREATE VIEW Bill_View AS
    -> SELECT S.bill_no, S.bill_date, S.cust_id,
    ->        S.item_id, I.price, S.qty_sold,
    ->        (I.price * S.qty_sold) AS amount
    -> FROM SALE S
    -> JOIN ITEM I ON S.item_id = I.item_id;
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE VIEW Weekly_Sales AS
    -> SELECT bill_date, SUM(I.price * S.qty_sold) AS Daily_Total
    -> FROM SALE S
    -> JOIN ITEM I ON S.item_id = I.item_id
    -> WHERE bill_date >= CURDATE() - INTERVAL 7 DAY
    -> GROUP BY bill_date;
Query OK, 0 rows affected (0.01 sec)

mysql>
```